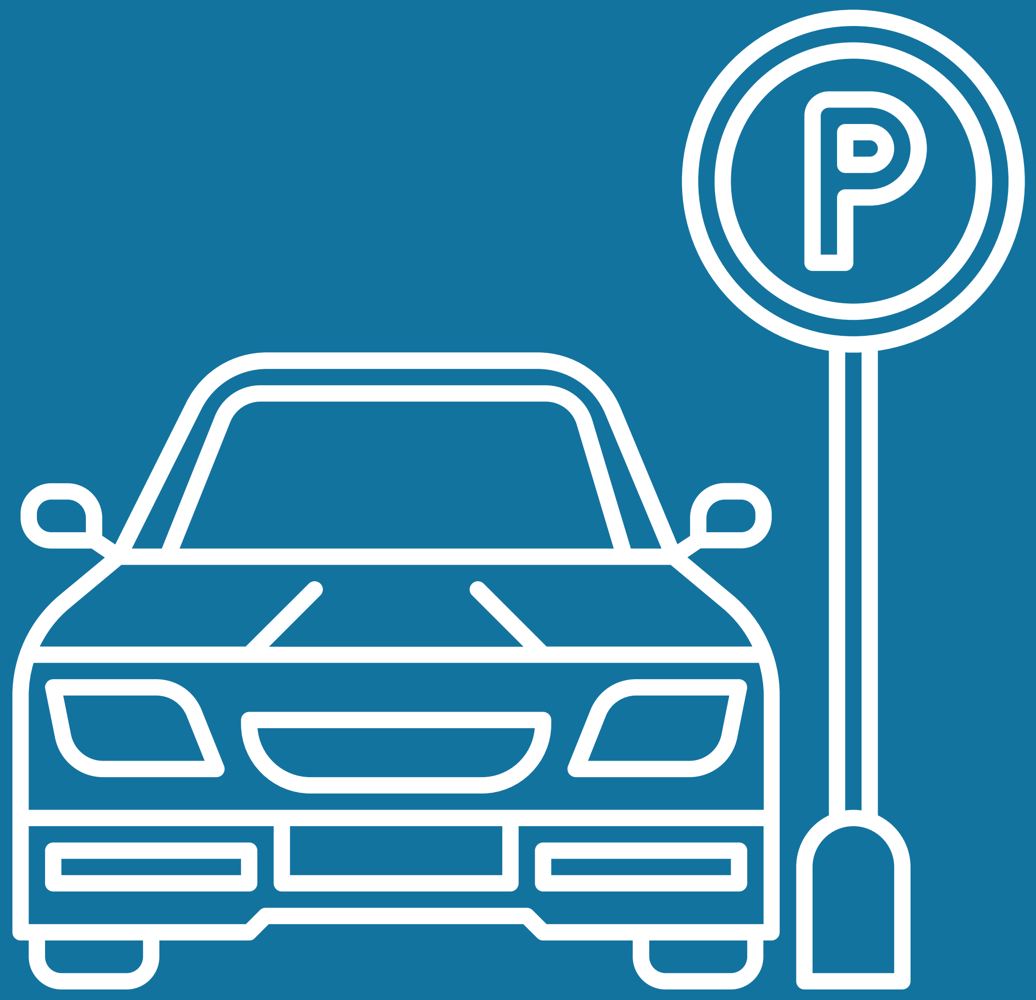
SECR3553

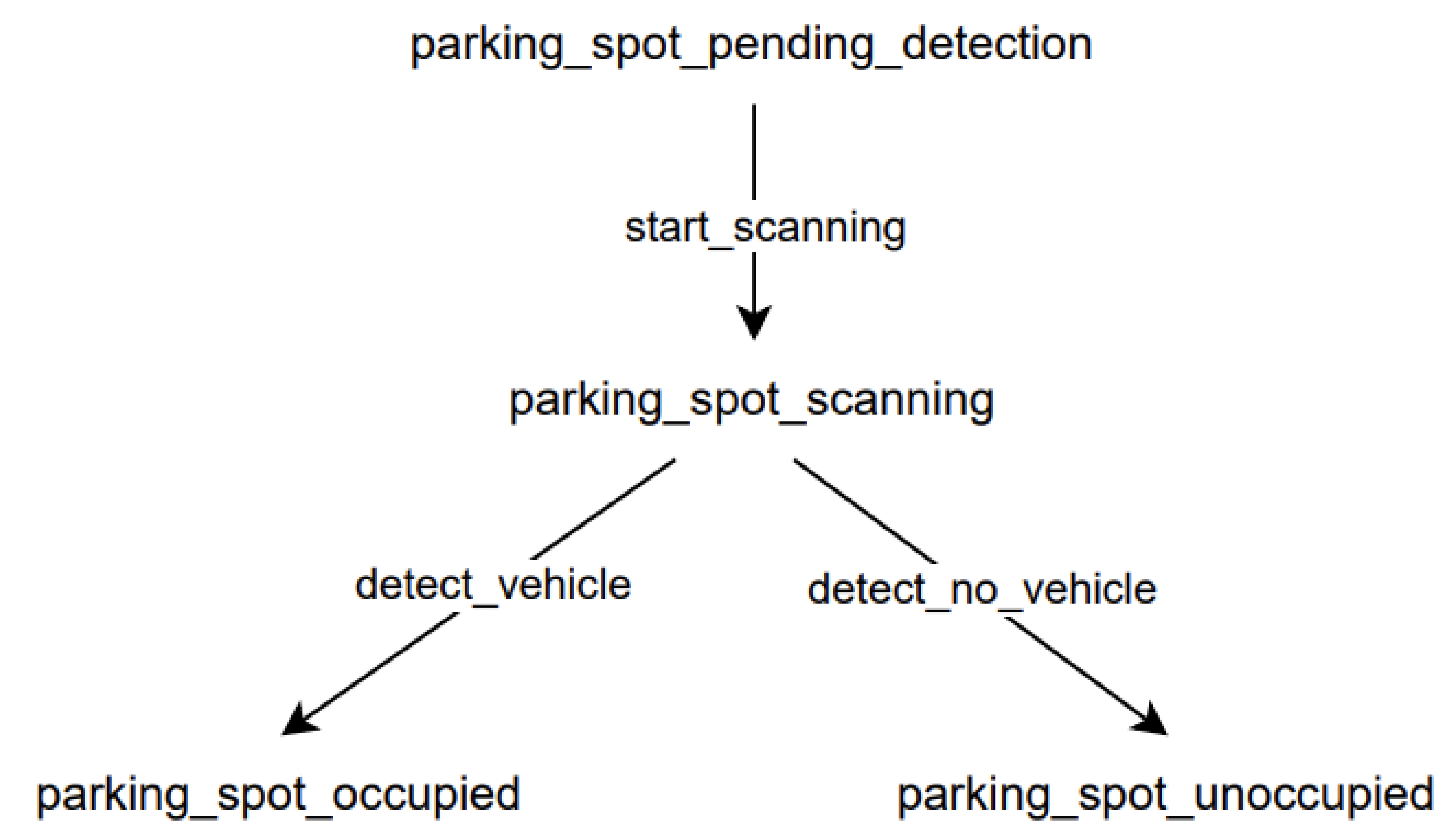
SMART CITY: PARKING DETECTION APPLICATION

AI Project Presentation
by Group 6

Introduction



- Parking congestion is a common problem in urban and campus areas
- Drivers waste time searching for available parking spaces
- Causes traffic congestion, fuel wastage and frustration
- Our project proposes an **AI-based Smart Parking Detection Application**
- Uses cameras and sensors to provide real-time parking availability



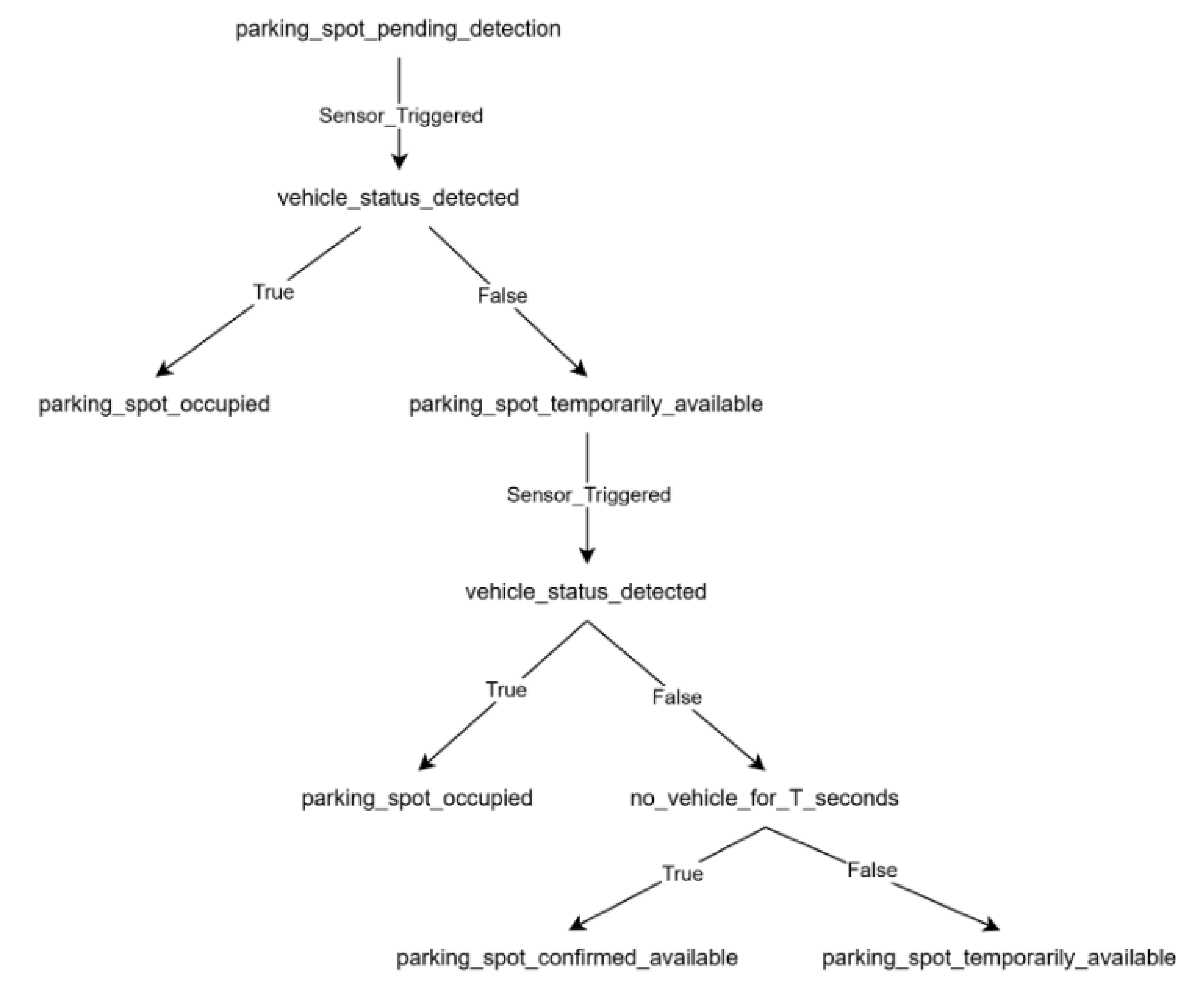
KR1: Detect Occupied Parking Space

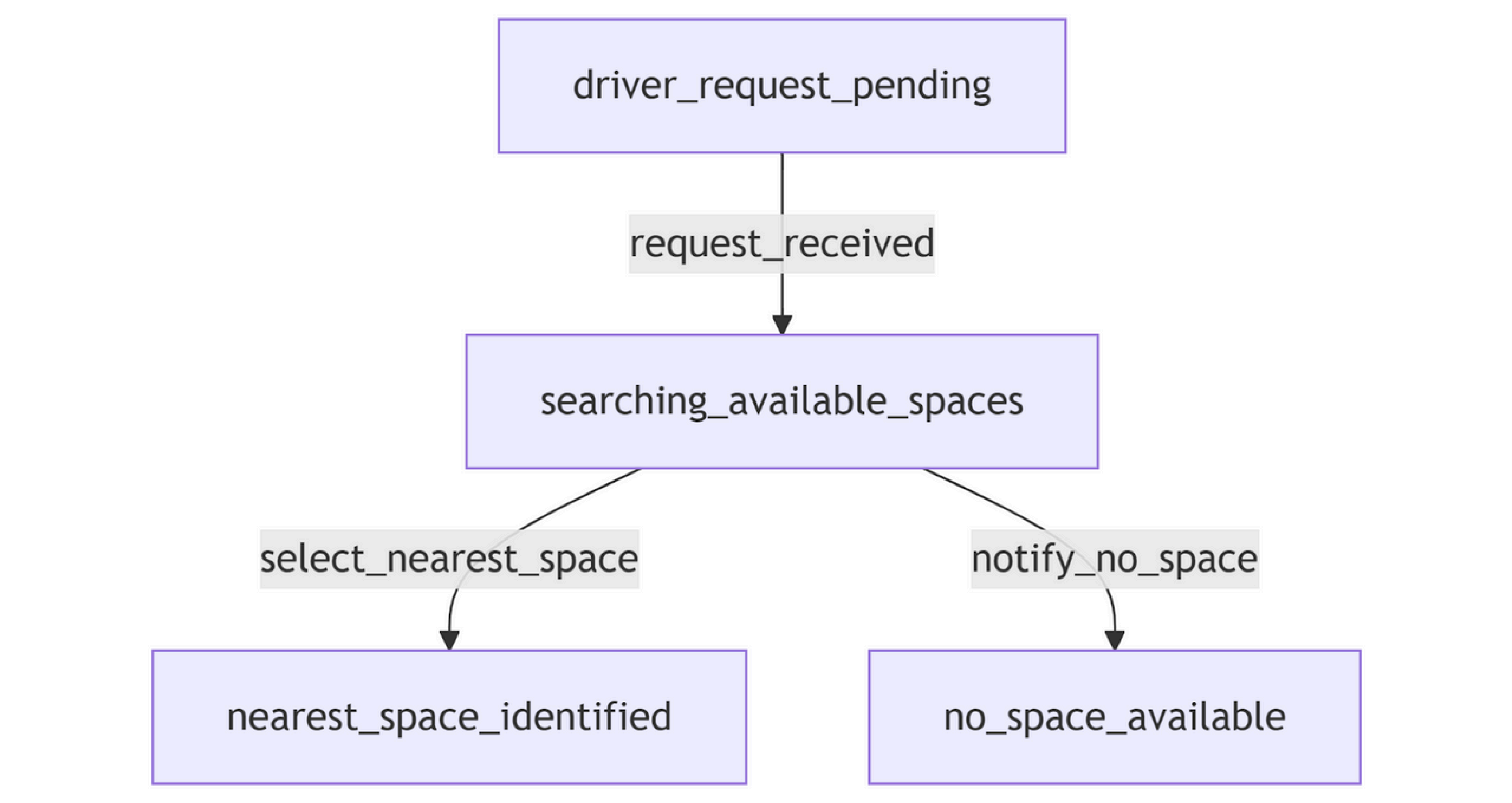
- IF vehicle_detected = TRUE THEN space_status = Occupied
- $\forall x\,(\mathrm{ParkingSpace}(x) \land \mathrm{VehicleDetected}(x) \rightarrow \mathrm{SpaceStatus}(x, \mathrm{Occupied}))$

KR2: Detect Available Parking Space

- IF vehicle_detected = FALSE THEN space_status = Available AND IF vehicle_detected = FALSE for a continuous period of T seconds THEN confirm space_status = Available

- ∀x (ParkingSpace(x) ∧ ¬VehicleDetected(x) → SpaceStatus(x, Available))
  ∀x∀t(ParkingSpace(x)∧NoVehicleDetectedFor(x,t,T)→ConfirmedStatus(x,Available))





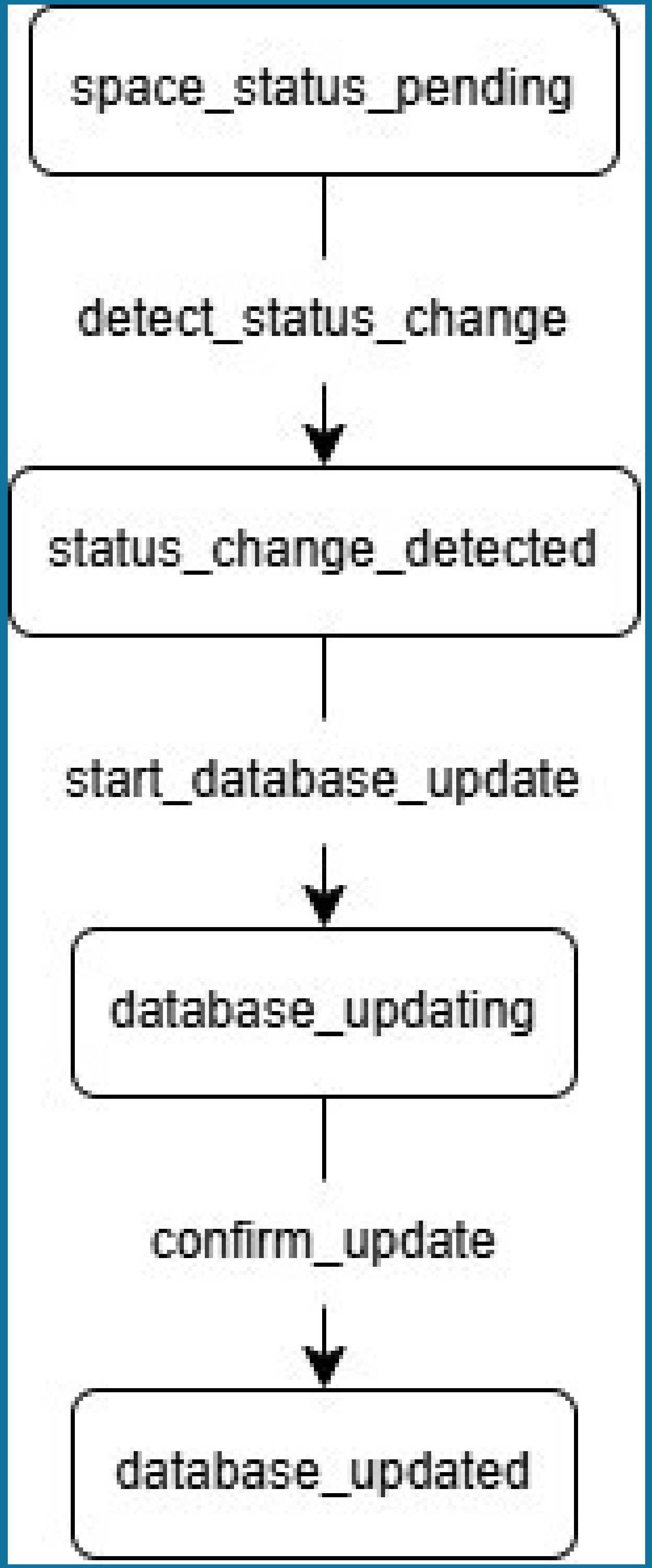
KR3: Provide Nearest Available Parking Spot

- IF driver_request = TRUE THEN display_nearest(space_status = Available)

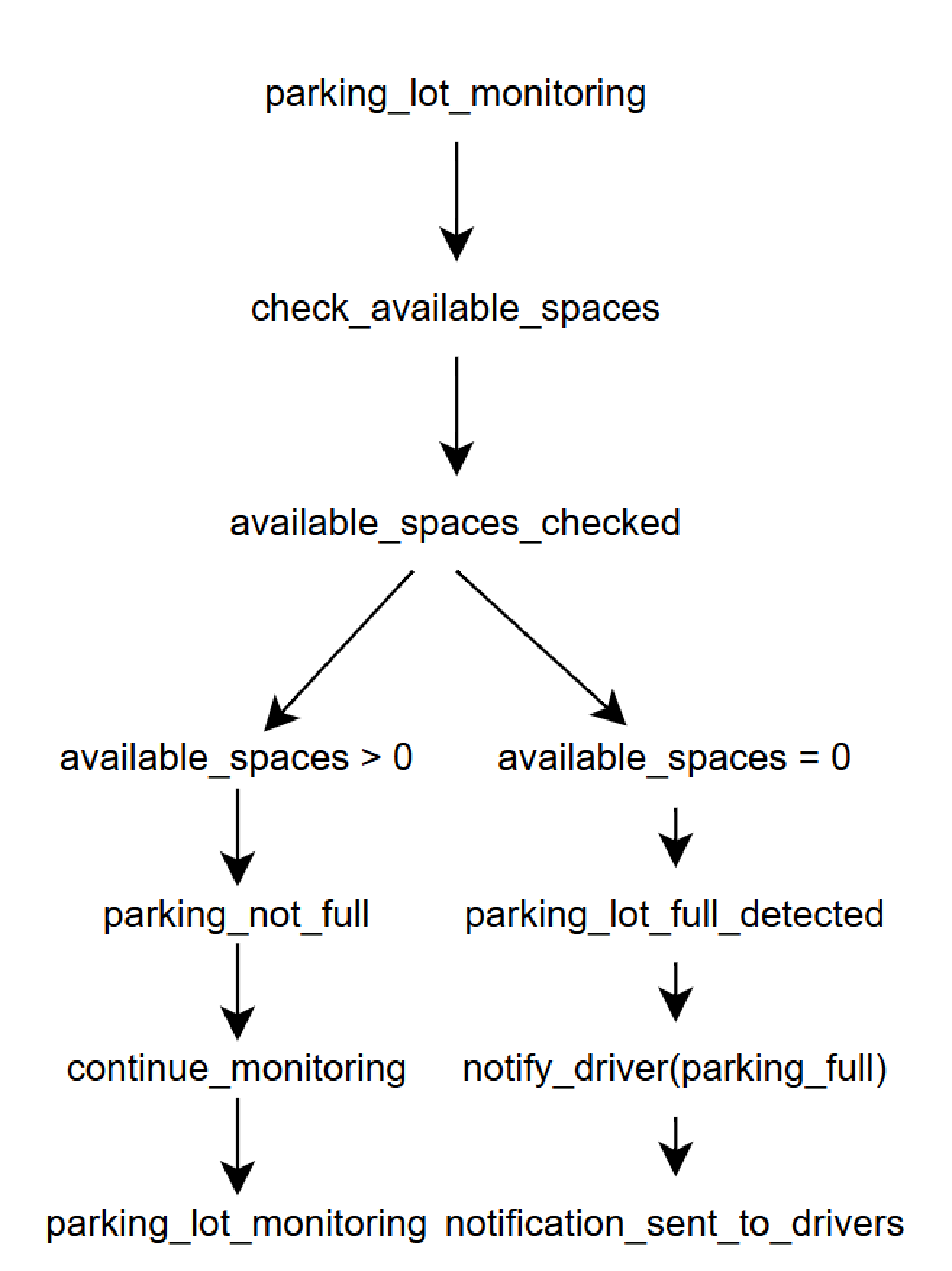
- $\forall x\ (DriverRequest(x) \rightarrow Display(NearestAvailableSpace(x)))$

KR4: Update Parking Database

- IF space_status changes THEN update database
- ∀x (ParkingSpace(x) ∧ StatusChanged(x) → UpdateDatabase(x))



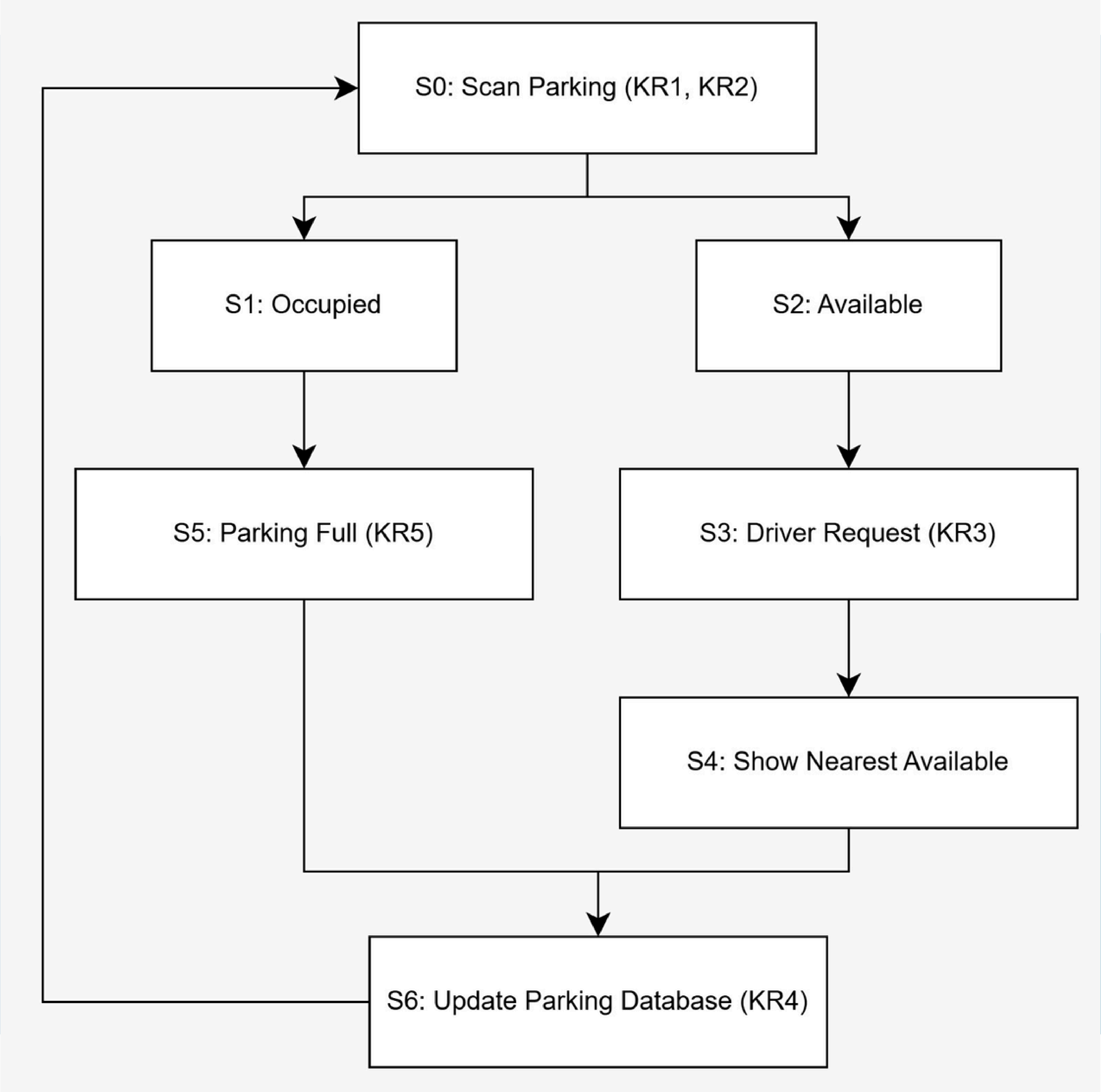
Smart city



KR5: Notify Driver When the Parking is Full

- IF available_spaces = 0 THEN notify_driver(parking_full)
- $\forall x\, (ParkingLot(x) \land AvailableSpaces(x) = 0 \rightarrow NotifyDriver(x, ParkingFull))$

Overall System State Space Graph



PEAS Model

Performance measure	- Accuracy of parking space detection - Speed of real-time database updates - Reduction in parking search time - Timely notifications when parking is full
Environment	- Cities or campuses parking areas - Parking spaces - Vehicles entering and leaving - Driver using the application - Cameras, sensors and network infrastructure
Actuators/Effectors	- Updating the parking database - Displaying available parking spaces on the application - Providing navigation guidance - Sending notifications to drivers when parking is full
Sensors	- Surveillance cameras - Parking occupancy sensors - GPS location data from drivers - User requests from the application - Database status logs

Thank You